4 Here are the first 4 terms of a sequence.

2 9 16 23

(a) (i) Write down the next term in the sequence.

23 + 7 = 30

1 mark
Continue the constant difference

30

(1)

(ii) Explain how you got your answer.

Add 7 to the previous term.

1 mark
State the term-to-term rule

(1)

(b) Work out the 10th term of the sequence.

Difference 7 gives 7n; 7 − 5 = 2, so the rule is 7n − 5.

n is the term position. Check n = 4: 7 × 4 − 5 = 23.

10th term = 7 × 10 − 5 = 65

1 mark
Use n = 10

65

(1)

(Total for Question 4 is 3 marks)